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The relationship between critical velocity, maximal lactate steady-state velocity and lactate turnpoint velocity in runners

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Abstract In cycle exercise, it has been suggested that critical power, maximal lactate steady state, and lactate turnpoint all demarcate the transition between the heavy exercise domain (in which blood lactate is elevated above resting values but remains stable over time) and the very heavy exercise domain (in which blood lactate increases continuously throughout constant-intensity exercise). The purpose of the present study was to assess the level of agreement between critical velocity (CV), maximal lactate steady-state velocity (MLSSV), and lactate turnpoint velocity (LTPV) during treadmill running. Eight male subjects [mean (SD) age 28 (5) years, body mass 71.2 (8.0) kg, maximum oxygen uptake 54.9 (3.2) ml·kg⁻¹·min⁻¹] performed an incremental treadmill test for the determination of LTPV (defined as a sudden and sustained increase in blood lactate concentration ([La]) at ≈ 2.0 –5.0 mM). The subjects returned to the laboratory on eight or nine occasions for the determination of CV and MLSSV. The CV was determined from four treadmill runs at velocities that were chosen to result in exhaustion within 2–12 min. The MLSSV was determined from four or five treadmill runs of up to 30 min duration and defined as the highest velocity at which blood [La] increased by no more than 1.0 mM after between 10 and 30 min of exercise. Analysis of variance revealed no significant differences between [mean (SD)] CV [14.4 (1.1) km·h⁻¹], MLSSV [13.8 (1.1) km·h⁻¹] and LTPV [13.7 (0.6) km·h⁻¹]. However, the bias $\pm 95\%$ limits of agreement for comparisons between CV and MLSSV [0.6 (2.2) km·h⁻¹], CV and LTPV [0.7 (2.7) km·h⁻¹], and MLSSV and LTPV [0.1 (1.8) km·h⁻¹] suggest that the extent of disagreement is too great to allow one variable to be estimated accurately from another in individual subjects. Direct

determination of MLSSV is necessary if precision is required in experimental studies.

Keywords Treadmill running · Exercise domains · Performance

Introduction

Whipp and Ward (1990) identified a number of exercise domains based on the respiratory and metabolic responses of subjects to exercise. At the onset of moderate exercise (below the lactate threshold, Th_{la}), oxygen uptake ($\dot{V}O_2$) rises in an exponential fashion, reaching a steady state within 2–3 min. In this domain, blood [La] may rise transiently as a result of the initial oxygen deficit before returning to resting levels as exercise proceeds. During heavy exercise (above the Th_{la} but below the critical power, $\dot{W}_{crit}$), $\dot{W}_{crit}$ rises exponentially to attain a delayed steady state that may be higher than the expected value, and blood [La] will be elevated above resting values but will not increase over time. During very heavy exercise (above the $\dot{W}_{crit}$ but below the maximum oxygen uptake, $\dot{V}O_{2max}$), $\dot{V}O_2$ and blood [La] do not attain a steady state, but continue to rise until the exercise is terminated. The continued rise of $\dot{V}O_2$ in the heavy and very heavy exercise domains to values that are greater than the anticipated steady-state values is brought about by the supplementation of the primary exponential response by the $\dot{V}O_2$ slow component. Finally, during severe exercise (short-term exercise above the $\dot{V}O_{2max}$), $\dot{V}O_2$ projects exponentially towards $\dot{V}O_{2max}$ and blood [La] rises steeply until exhaustion occurs. According to Whipp and Ward (1990), the transitions between the various exercise domains for cycle exercise are demarcated by the Th_{la} (for moderate to heavy exercise), the $\dot{W}_{crit}$ (for heavy to very heavy exercise) and the $\dot{V}O_{2max}$ (for very heavy to severe exercise), and these variables have been shown to be useful for assessing exercise tolerance, prescribing training, and for predicting performance (Jones and Carter 2000).

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The $\dot{W}_{\text{crit}}$ concept was first described by Monod and Scherrer (1965), who noted a hyperbolic relationship between power output and time to exhaustion during exercise in small muscle groups. Following transformation, the slope of the regression line relating total work done to time to exhaustion was defined as the $\dot{W}_{\text{crit}}$ (i.e. the power output that can be sustained for a long time without fatigue), while the intercept was considered to represent a finite energy store (Monod and Scherrer 1965). Moritani et al. (1981) extended the $\dot{W}_{\text{crit}}$ concept to whole-body exercise. They proposed that cycling at a power output below the $\dot{W}_{\text{crit}}$ could theoretically be maintained indefinitely (although realistically, it is likely that fatigue would eventually result from glycogen depletion or dehydration), whereas cycling at a power output above $\dot{W}_{\text{crit}}$ would result in the accumulation of blood lactate and depletion of the stored energy sources at a predictable rate until exhaustion. Poole et al. (1988) demonstrated that constant-load cycle exercise below the $\dot{W}_{\text{crit}}$ resulted in stable $\dot{V}\text{O}_2$ and blood [La] values over time, while exercise just above the $\dot{W}_{\text{crit}}$ caused $\dot{V}\text{O}_2$ and blood [La] to increase with time until the exercise was terminated. Therefore, it appears that the $\dot{W}_{\text{crit}}$ may demarcate the transition from heavy exercise to very heavy exercise, and provide a noninvasive measure of the maximal lactate steady state (MLSS); that is, the highest exercise intensity at which blood [La] (and $\dot{V}\text{O}_2$) can be stabilised during long-term exercise (Jones and Doust 1998). The MLSS is determined from the response of blood lactate to a series (≈ 4 –5) of constant-intensity exercise bouts of up to 30 min duration, and is defined as the highest power output or velocity at which blood [La] increases by no more than 1.0 mM after between 10 and 30 min of exercise (Beneke and von Duvillard 1996; Jones and Doust 1998).

Previous studies have assessed the agreement between $\dot{W}_{\text{crit}}$ and the MLSS by measuring the blood [La] response to exercise at power outputs equivalent to, or just above and just below, the pre-determined $\dot{W}_{\text{crit}}$ (Housh et al. 1989; Jenkins and Quigley 1990; McLellan and Cheung 1992; Poole et al. 1988; Wakayoshi et al. 1993). Some of these studies suggest that the $\dot{W}_{\text{crit}}$ and MLSS are identical (Poole et al. 1988; Vautier et al. 1995; Wakayoshi et al. 1993), while others suggest that the $\dot{W}_{\text{crit}}$ overestimates MLSS and that $\dot{W}_{\text{crit}}$ cannot be sustained beyond ≈ 20 –33 min without fatigue (Housh et al. 1989; Jenkins and Quigley 1990; McLellan and Cheung 1992). Interestingly, no previous study has compared $\dot{W}_{\text{crit}}$ with the directly and independently determined MLSS. Furthermore, the relationship between critical velocity (CV) and maximal lactate steady state velocity (MLSSV) has not been fully explored in other modes of exercise such as running.

Unlike the Th_{la} and the $\dot{V}\text{O}_{2\text{max}}$, which can be determined from a single incremental exercise test, the determination of $\dot{W}_{\text{crit}}$ /CV requires the subject to perform 3–6 constant-load exercise bouts to exhaustion (Hill 1993). Likewise, determination of the MLSSV requires the subject to perform 4–5 exercise bouts of 30 min

duration, preferably on separate days (Beneke and von Duvillard 1996; Jones and Doust 1998). These procedures are time-consuming and demanding both for the subject and the experimenter. Recently, it has been suggested that the exercise intensity at the lactate turnpoint (defined as the second threshold increase in blood [La] at ≈ 2.0 –5.0 mM during incremental exercise) provides a good estimate of the exercise intensity at the MLSS (Aunola and Rusko 1992; Hofmann et al. 1994). If this can be verified, it would offer an alternative to the direct determination of $\dot{W}_{\text{crit}}$ /CV and MLSS.

Given the limitations to the existing research highlighted above, the purpose of the present study was to assess the relationship between CV, MLSSV and the lactate turnpoint velocity (LTPV) during treadmill running.

Methods

Subjects

Eight male subjects [mean (SD) age 28 (5) years, body mass 71.2 (8.0) kg, body fat 13.4 (3.4)%, $\dot{V}\text{O}_{2\text{max}}$ 54.9 (3.2) ml·kg⁻¹·min⁻¹] gave written informed consent to participate in this study, which was approved by the Institutional Ethics Committee. The subjects were active in recreational sports activities, and they were familiar with treadmill running, exercise physiology laboratory procedures and exercising to exhaustion.

Experimental design and general procedures

The subjects completed a total of nine laboratory sessions: an initial incremental treadmill test to determine Th_{la} , LTPV, and $\dot{V}\text{O}_{2\text{max}}$, and eight subsequent sessions to determine CV and MLSSV. The CV and MLSSV testing sessions were presented in blind random order and completed on separate days within a 3-week period. The subjects were instructed to arrive at the laboratory in a rested and fully hydrated state, at least 3 h post-prandial, and to avoid strenuous exercise in the 48 h preceding a test session. They wore the same running shoes and lightweight running kit for all tests. For each subject, the tests took place at the same time of day (± 2 h) to minimise the effects of diurnal biological variation. Before each of the tests, the subjects completed a warm-up of 5–10 min of self-paced jogging and some stretching exercises.

All treadmill tests took place on a Powerjog EG20 with the treadmill grade set at 1%. During all tests, the treadmill output display was hidden from the subjects' view to prevent them seeing the treadmill belt velocity and the distance covered. Respiratory variables were determined by collection of expired air into Douglas bags over a timed period. The expired air was immediately analysed for the concentrations of O_2 and CO_2 by sampling through electronic gas analysers (Servomex Series 1400, Crowborough, UK), which were calibrated before each test using certified standard gases (British Oxygen, London, UK). Volumes were determined by a calibrated dry gas meter (Harvard, Edenbridge, UK). Heart rate (f_c) was determined by telemetry (Polar Electro Oy, Kempele, Finland). Capillary blood samples were taken when appropriate (see below) during short (≈ 15 s) interruptions in the exercise. The fingertip was cleaned with an alcohol swab, dried and then punctured with an automated lancet before ≈ 60 μl of blood was collected into a capillary tube. Blood samples were analysed for [La] in duplicate using an automated analyser (YSI 1500 Sport, Yellow Springs, Ohio, USA). The coefficient of variation for blood [La] determination was 1.1% for ten samples in the physiological range (5 mM).

Determination of lactate threshold velocity, LTPV and $\dot{V}O_{2\max}$

On arrival at the laboratory, each subject's height and body mass were recorded. The subjects then completed a multi-stage treadmill test for assessment of the blood lactate response to submaximal exercise and for determination of lactate threshold velocity (LTV) and LTPV. The test protocol involved 8–10 submaximal stages of 4-min duration. The initial velocity was set following discussion with the subject (range 8–10 km·h⁻¹) and running velocity was increased by 1.0 km·h⁻¹ at the completion of each stage. Expired air was collected into a Douglas bag over a timed period within the last 60 s of each stage, and the average f_c over the last 30 s of each stage was recorded. Fingertip capillary blood samples were collected during brief (≈ 15 s) pauses between exercise stages and analysed for blood [La] (see above). The submaximal test protocol was terminated when subjects exceeded 95% of maximal f_c ($f_{c,\max}$; this was known from previous experience in the laboratory). At this point, the treadmill velocity was reduced by 2.0 km·h⁻¹ and the treadmill grade was increased by 1% every minute until the subject became exhausted (within 6–8 min). During this period, expired air was collected into Douglas bags over the last 40–50 s of each minute. The highest $\dot{V}O_2$ that was recorded was accepted as $\dot{V}O_{2\max}$. At this point (i.e. at exhaustion), a plateau in $\dot{V}O_2$ was apparent and the respiratory exchange ratio exceeded 1.10 in all subjects.

Coded plots of blood [La] against running speed were presented to two independent reviewers for assessment of LTV and LTPV. The LTV was defined as the final running velocity before the first sustained increase in blood [La] above baseline (Beaver et al. 1985). Following determination of LTV by the reviewers, the blood [La] data before LTV were deleted (Hofmann et al. 1994), and log-log plots of the remaining blood [La] values against running velocity were re-presented to the reviewers for determination of the LTPV. The LTPV was defined as the final running velocity before the observation of a sudden and sustained increase in blood [La]. By definition, this LTPV must lie between Th_{la} and $\dot{V}O_{2\max}$, and it therefore represents the “second lactate threshold” at ≈ 2 –5 mM identified by other authors (Aunola and Rusko 1992; Skinner and McLellan 1980). The inter-reviewer reliability for plot assessment was perfect ($r = 1.0$).

Determination of CV

CV was determined using the procedures of Hughson et al. (1984). All subjects performed four randomly assigned treadmill runs to exhaustion at velocities ranging from 14.6 to 20.8 km·h⁻¹. The subjects ran at 100, 105, 110 and 120% of their estimated running velocity at $\dot{V}O_{2\max}$ ($v \cdot \dot{V}O_{2\max}$). The $v \cdot \dot{V}O_{2\max}$ was determined by solving individual regression equations relating $\dot{V}O_2$ to submaximal running velocity for $\dot{V}O_{2\max}$. The subjects stood with their feet astride the treadmill belt, which was pre-set to the required velocity, and the timing for each bout began when the subjects let go of the treadmill guardrails and began running. The bout was terminated when the subjects grasped the guardrails, signalling exhaustion, and time to exhaustion was recorded to the nearest second. The subjects were given strong verbal encouragement throughout the runs. f_c was monitored throughout each of the runs and was recorded at exhaustion. The CV was calculated as the slope of the regression line relating total distance covered at each of the treadmill velocities to the times to exhaustion at those velocities.

Determination of MLSSV

Subjects completed four treadmill runs for the determination of MLSSV. The runs were of 30 min duration unless subjects fatigued before this time. The running velocities were individually selected according to ability, but were within the range of 0.5–3.0 km·h⁻¹ above the LTV. The velocity of the runs varied by 0.5 km·h⁻¹. Blood [La] was determined at rest and every 5 min during exercise until 30 min had elapsed or exhaustion occurred. Expired air was

collected at rest and between minutes 4–5, 9–10, 14–15, 19–20, 24–25, and 29–30. f_c was recorded throughout the exercise bouts. Blood [La] was plotted against time for each of the running velocities, and MLSSV was defined as the highest velocity that produced an increase in blood [La] of no more than a 1.0 mM after between 10 and 30 min of exercise (Beneke and von Duvillard 1996; Jones and Doust 1998).

Statistical analysis

Data are reported as mean (SD) unless indicated otherwise. Analysis of variance with repeated measures was used to test for the existence of significant differences between CV, MLSSV and LTPV. The relationship between these variables was examined using Pearson product moment correlation coefficients. In addition, we calculated the bias and limits of agreement (Bland and Altman 1986) between the three variables.

Results

For all subjects, the runs for the determination of CV resulted in exhaustion in between 2 and 12 min. The mean (SD) times to exhaustion were 531 (84) s, 515 (131) s, 354 (196) s and 252 (63) s at 100%, 105%, 110% and 120% of the $v \cdot \dot{V}O_{2\max}$, respectively. For each of the runs, the f_c at exhaustion [190 (9) beats·min⁻¹] was not significantly different from the $f_{c,\max}$ measured during the $\dot{V}O_{2\max}$ test [190 (8) beats·min⁻¹]. Figure 1 demonstrates the derivation of CV in a typical subject. The model used for derivation of CV (distance covered against time to exhaustion) provided an excellent fit to the data ($R^2 = 0.991$ – 0.999). The MLSSV was determined in all subjects. Figure 2 demonstrates the response of blood [La] over time at a range of running velocities in a typical subject, and highlights the determination of the MLSSV. Exercise above the MLSSV resulted in a continuous increase in blood [La] until the exercise was terminated in all subjects. The LTPV was determined in all subjects using log-log plots of blood [La] against running velocity for the region between Th_{la} and $\dot{V}O_{2\max}$. A clear breakpoint in blood [La] was apparent in all subjects.

Analysis of variance revealed no significant differences between CV [14.4 (1.1) km·h⁻¹], MLSSV [13.8 (1.1) km·h⁻¹] and LTPV [13.7 (0.6) km·h⁻¹]. However, due to the relatively homogeneous fitness of the subjects, the three variables were not significantly correlated (Fig. 3). Furthermore, there were no significant differences between the estimated $\dot{V}O_2$ at CV [3.50 (0.50) l·min⁻¹; 89 (4)% $\dot{V}O_{2\max}$], the $\dot{V}O_2$ at MLSSV [3.39 (0.53) l·min⁻¹; 86 (4)% $\dot{V}O_{2\max}$] and the $\dot{V}O_2$ at LTPV [3.30 (0.46) l·min⁻¹; 84 (3)% $\dot{V}O_{2\max}$]. There were no significant differences between the blood [La] at MLSSV [2.2 (0.1) mM] and LTPV [2.2 (0.8) mM], or the f_c at MLSSV [172 (6) beats·min⁻¹; $\approx 91\%$ of $f_{c,\max}$] and LTPV [175 (12) beats·min⁻¹; $\approx 92\%$ of $f_{c,\max}$].

However, despite the lack of significant differences between the group mean responses, the bias $\pm 95\%$ limits of agreement for comparisons between CV and MLSSV [0.6 (2.2) km·h⁻¹], CV and LTPV [0.7 (2.7) km·h⁻¹], and MLSSV and LTPV [0.1

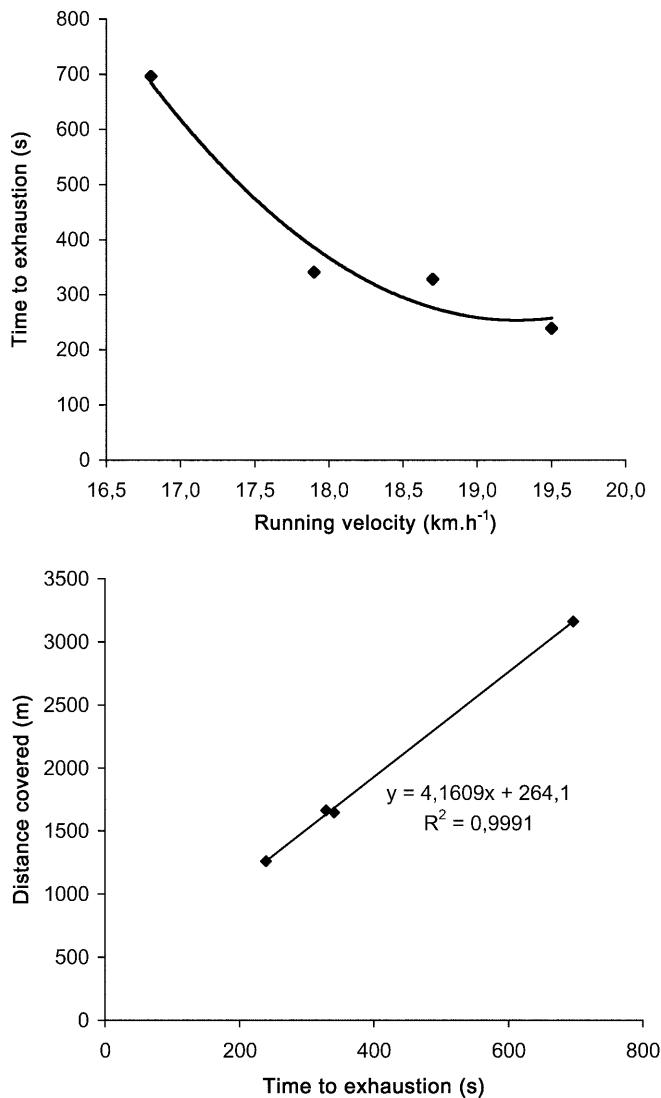


Fig. 1 Derivation of the critical velocity (CV) in a typical subject. The hyperbolic relationship between time to exhaustion and running velocity is transformed into a linear relationship between distance covered and time to exhaustion. In this model, the slope of the regression equation is the CV

(1.8) $\text{km}\cdot\text{h}^{-1}$] suggest that the extent of disagreement is too great to allow one variable to be accurately estimated from another in individual subjects (Fig. 4). The relationships between the running velocities at Th_{la} , LTP, MLSS, CV and $\dot{V}\text{O}_{2\text{max}}$ for individual subjects are shown in Fig 5.

Discussion

Conventional statistical approaches demonstrated that there was no significant difference between CV, MLSSV, and LTPV. This suggests that CV, MLSSV, and LTPV could be used interchangeably to demarcate the boundary between the heavy and very heavy exercise intensity domains. However, while this might be true on

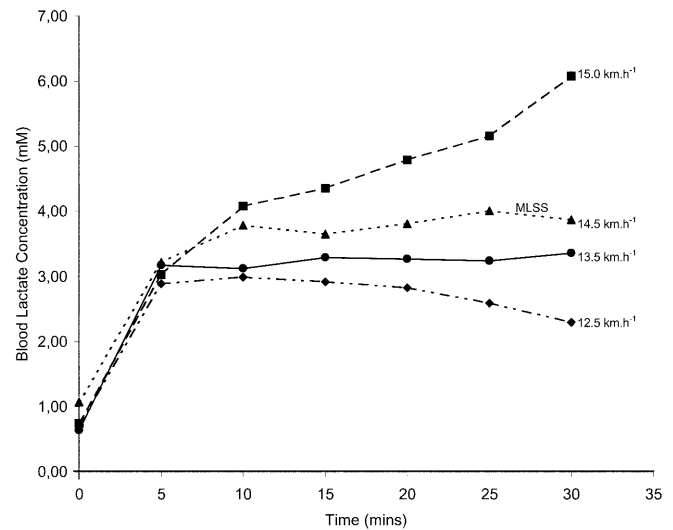


Fig. 2 Determination of the maximal lactate steady state velocity (MLSSV) in a typical subject. Subjects complete a series of constant-velocity treadmill runs and the highest velocity that allows a stabilisation of blood lactate concentration over time (less than a 1.0-mM increase between 10 and 30 min of exercise) is defined as the MLSSV

a conceptual level when mean values are considered, the bias $\pm 95\%$ limits of agreement (Bland and Altman 1986) indicate that estimating one variable from another in individual subjects might result in significant error. For example, if the MLSSV were to be estimated from the LTPV in a single subject, the 95% limits of agreement suggest that the MLSSV might be as much as 2.1 $\text{km}\cdot\text{h}^{-1}$ above LTPV, or as much as 1.5 $\text{km}\cdot\text{h}^{-1}$ below LTPV. The potential for error is obviously too great and such estimates would be unwise if precision were needed in experimental studies.

Hill (1927) first demonstrated that a runner's performance could be described by the relationship between running velocity and the time to exhaustion at that velocity. Hughson et al. (1984) were the first to apply this "critical velocity" (CV) concept to treadmill running. The CV can be determined as the slope of the regression equation relating distance covered to time to exhaustion, or as the intercept of the regression equation relating velocity to the inverse of time to exhaustion. It should be noted that these models should only be applied in instances where the energy cost per unit distance of locomotion is independent of the velocity. While this is essentially true for running, it is not true for other forms of locomotion such as cycling and swimming (di Prampero 1999). The $\dot{W}_{\text{crit}}/\text{CV}$ relationship has been shown to correlate with $\dot{V}\text{O}_{2\text{max}}$ (Housh et al. 1991; Hughson et al. 1984; Pepper et al. 1992), a variety of lactate and ventilatory thresholds (McLellan and Cheung 1992; Moritani et al. 1981; Wakayoshi et al. 1993), electromyographic fatigue threshold (de Vries et al. 1982; Le Chevalier et al. 2000) and endurance exercise performance (Hughson et al. 1984; McLellan and Cheung 1992; Moritani et al. 1981; Poole et al. 1988;

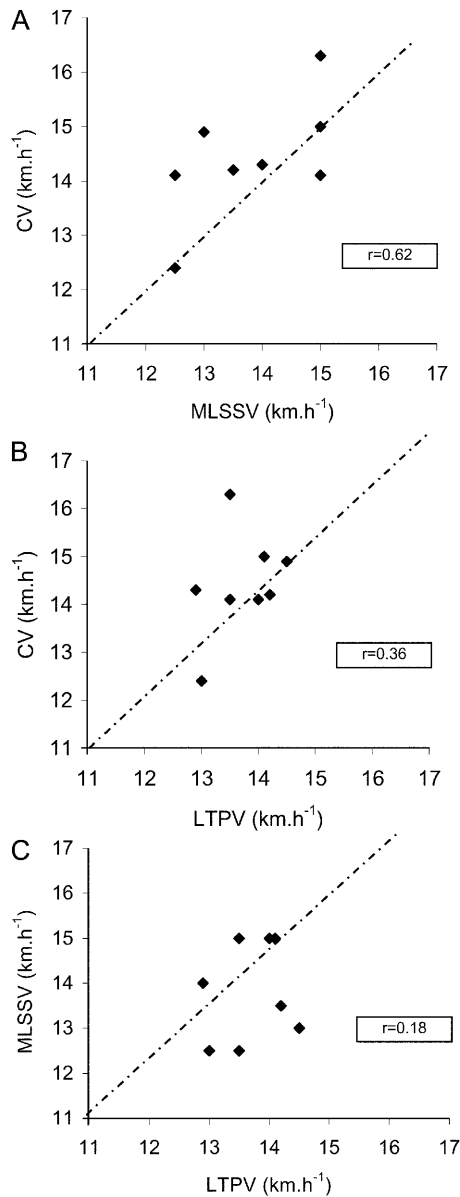


Fig. 3 Pearson product moment correlation coefficients between CV and MLSSV (A), CV and lactate turnpoint velocity (*LTPV*; B), and MLSSV and LTPV (C). The *dashed line* is the line of identity

Wakayoshi et al. 1993), and to be sensitive to endurance training (Jenkins and Quigley 1992; Poole et al. 1990). The primary disadvantage of the measurement of $\dot{W}_{crit}/CV$ in practice is the requirement for subjects to perform a number (3–6) of exercise bouts to complete exhaustion over several days (Hill 1993). This has led investigators to explore ways to estimate the $\dot{W}_{crit}/CV$ relationship and to reduce the number of exhaustive exercise bouts required for its determination (Housh et al. 1990; Kranenburg and Smith 1996; Wakayoshi et al. 1992).

The study presented here is the first to compare CV to the directly and independently determined MLSSV. Our group mean results indicate that CV and MLSSV probably represent the same physiological phenomenon,

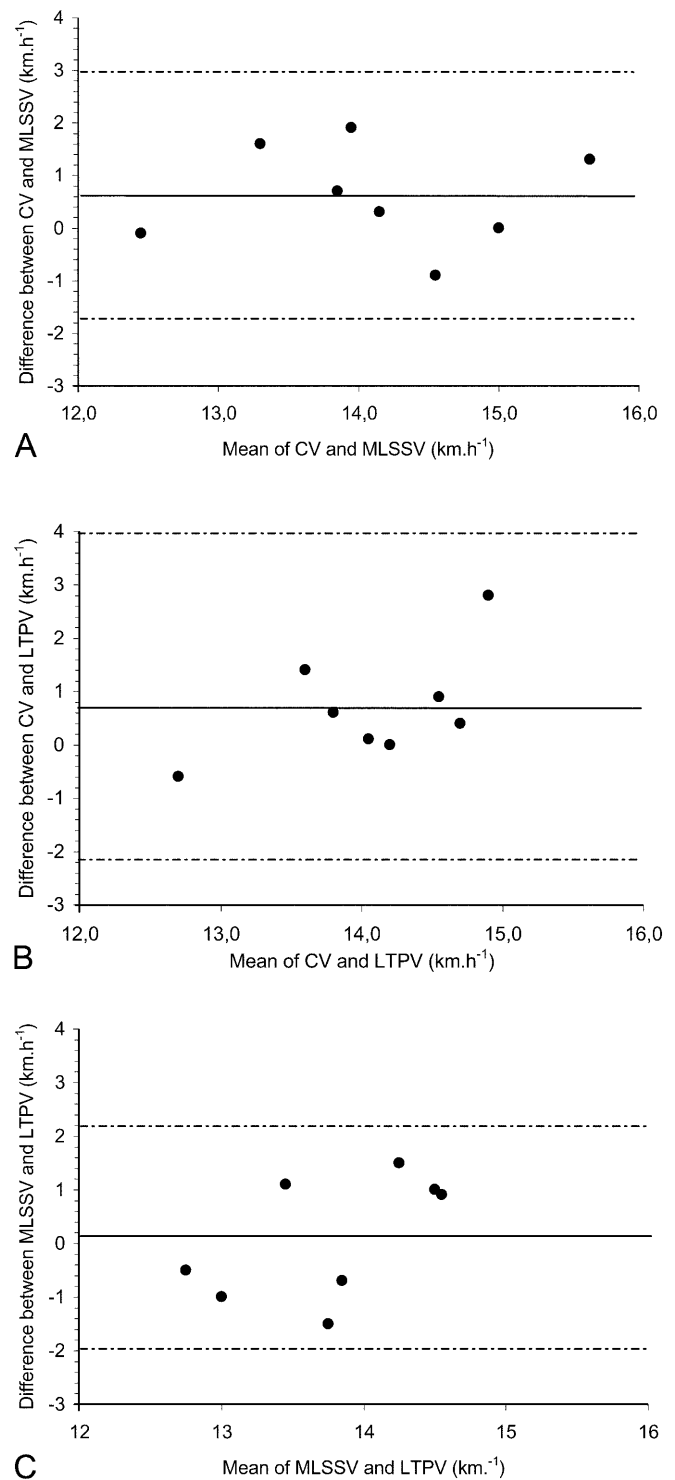
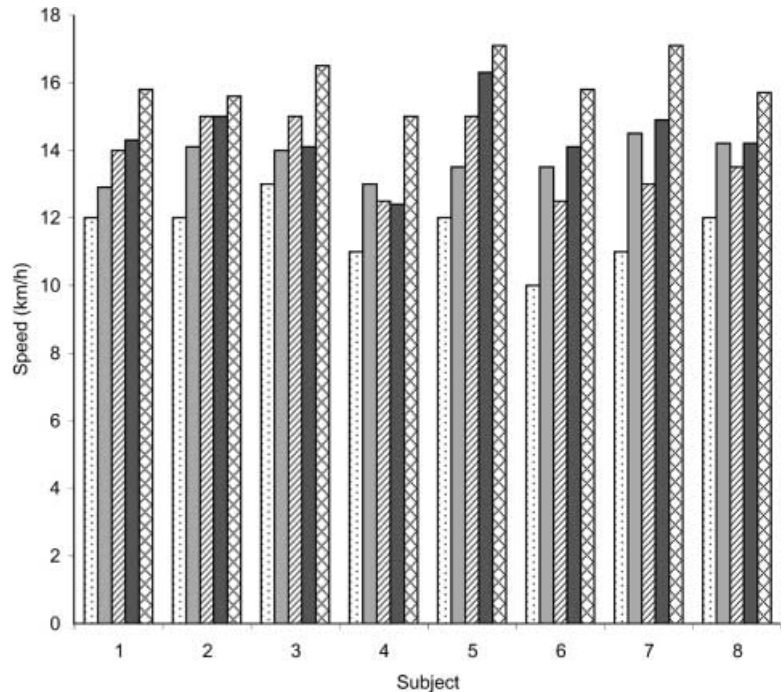


Fig. 4 Bland-Altman plots showing the bias and limits of agreement in comparisons of CV and MLSSV (A), CV and LTPV (B), and MLSSV and LTPV (C). The *solid horizontal line* represents the bias between the two measures (i.e. the mean difference between the group means for the two variables). The *dashed lines* represent the 95% limits of agreement between the two variables, and reflect the extent to which one variable might be expected to differ from another in individual subjects

Fig. 5 Representation of the individual values of lactate threshold velocity (*spotted bar*), LTPV (*lightly shaded bar*), MLSSV (*diagonal lined bar*), CV (*dark shaded bar*) and estimated velocity at maximum oxygen uptake (*cross-hatched bar*)



although it should be cautioned that the error in measuring CV and/or MLSSV limits the accuracy with which one variable may be estimated from the other in individual subjects. Other studies have also shown agreement between $\dot{W}_{\text{crit}}/\text{CV}$ and MLSS in swimming (Wakayoshi et al. 1992), cycling (Poole et al. 1988; Vautier et al. 1995) and running (Sid-Ali et al. 1991). In these studies, however, the proximity to MLSS was estimated based on the response of blood [La] to constant-intensity exercise above and below the pre-determined $\dot{W}_{\text{crit}}/\text{CV}$. It should be noted that in our study, the CV was higher than the MLSSV in five of the eight subjects. While this may be a function of the fact the MLSSV was only measured to the nearest $0.5 \text{ km}\cdot\text{h}^{-1}$, it is also possible that there is a tendency for CV to slightly overestimate MLSSV. The CV and VMLSSV demanded $\approx 89\%$ and $\approx 86\%$ of $\dot{V}\text{O}_{2\text{max}}$, respectively, and the mean difference between CV and MLSSV was $0.6 \text{ km}\cdot\text{h}^{-1}$. Running at only $0.5 \text{ km}\cdot\text{h}^{-1}$ above the MLSSV can result in significant blood lactate accumulation over time (Fig. 2). This supports some previous studies in which it was shown that exercise at the $\dot{W}_{\text{crit}}$ cannot be sustained without a significant accumulation of blood lactate and fatigue within 20–33 min on average (Housh et al. 1989; Jenkins and Quigley 1990; McLellan and Cheung 1992).

Although we did not measure the physiological responses of our subjects during exercise at the CV, previous studies have suggested that CV overestimates endurance running performance. Hughson et al. (1984) estimated 10-km race times of subjects from their CV and found that the predicted times were 2–3 min faster than the actual times. Pepper et al. (1992) reported that time to exhaustion at 100% of CV was only 16.4 (6.1) min, while Housh et al. (1991) estimated that exercise at CV would result in exhaustion in 9.6–16.8 min.

Interestingly, in all of these studies (Hughson et al. 1984; Housh et al. 1991; Pepper et al. 1992), CV was defined as the intercept of the velocity/time relationship. Recently, Kranenburg and Smith (1996) reported that CV determined from track and treadmill tests was highly correlated with 10-km race velocity and that the mean values for CV and 10-km race velocity were essentially the same.

It has been suggested that methodological limitations increase the potential for $\dot{W}_{\text{crit}}/\text{CV}$ to overestimate the MLSS (Hill 1993). In deciding upon which exercise intensities (and hence the exercise duration) to use for $\dot{W}_{\text{crit}}/\text{CV}$ determination, the experimenter has to reach a compromise between choosing bouts that are so short that they may be unrepresentative of the demands of endurance exercise and that may contravene the assumption of a hyperbolic relationship between power output and time to exhaustion, and bouts that are so long that psychological factors contribute to the decision to terminate exercise (Bishop et al. 1998; Hill 1993). Another consideration is the number of exercise bouts used to calculate $\dot{W}_{\text{crit}}$; too many trials reduce the practicality of the test while too few trials reduce the accuracy of $\dot{W}_{\text{crit}}/\text{CV}$ determination (Hill 1993). The consensus of opinion is that 3–4 exercise bouts of between 2 and 15 min duration, as used in the present study, may be optimal for $\dot{W}_{\text{crit}}/\text{CV}$ determination (Hill 1993; Housh et al. 1990). We are confident that our subjects gave maximal effort during the runs used to determine CV. This is supported by the fact that the terminal f_c for each run was equal to the $f_{c,\text{max}}$ measured during the $\dot{V}\text{O}_{2\text{max}}$ test. Furthermore, the times to exhaustion at 100, 105, 110, and 120% $v\cdot\dot{V}\text{O}_{2\text{max}}$ are similar to those reported previously (Billat et al. 1994, 1995). We chose to estimate CV by calculating the slope

of the regression line relating total distance covered to time to exhaustion, since it has been suggested that the slope of the regression line is less prone to error than is the y-intercept (Vandewalle et al. 1997), and because this model produced an excellent fit to the data ($R^2 = 0.991-0.999$). Kranenburg and Smith (1996) reported that CV was higher in 14 out of 18 subjects when CV was calculated as the y-intercept of the regression equation relating running velocity to time, than when it was calculated using the distance covered/time to exhaustion model. Recently, it has been suggested that modelling the untransformed nonlinear data might overcome two questionable assumptions inherent in the $\dot{W}_{\text{crit}}$ concept: (1) that power is infinite as time to exhaustion approaches zero, and (2) that the energy stores represented by the “anaerobic work capacity” term are totally depleted at exhaustion (Gaesser et al. 1995; Morton 1996). This approach appears to result in lower estimates of $\dot{W}_{\text{crit}}$ than do conventional models (Bull et al. 2000). In the context of the present study, it can be stated that there was no significant mean difference between CV and MLSSV, despite the fact that we used a linear model that may have slightly overestimated the CV.

The MLSSV is defined as the highest running velocity at which a balance exists between the entry of lactate into the blood and its removal from the blood during constant-intensity exercise of approximately 30 min duration (Beneke and von Duvillard 1996; Jones and Doust 1998). We determined the MLSSV from four treadmill runs of up to 30 min duration at running velocities that were between $0.5 \text{ km}\cdot\text{h}^{-1}$ and $3.0 \text{ km}\cdot\text{h}^{-1}$ higher than the Th_{la} . Jones and Doust (1998) reported that the MLSSV occurred at $0.5-1.0 \text{ km}\cdot\text{h}^{-1}$ above the Th_{la} in well-trained runners. In the present study, the MLSSV occurred at $\approx 2.2 \text{ km}\cdot\text{h}^{-1}$ above the LT and $\approx 2.3 \text{ km}\cdot\text{h}^{-1}$ below the estimated $v\text{-}\dot{V}\text{O}_{2\text{max}}$ (Fig. 5). In order to normalise exercise intensity between individuals or exercise modalities, it is common to exercise subjects at calculated percentages of the difference between the velocity at which the Th_{la} occurs and $\dot{V}\text{O}_{2\text{max}}$, using the “% delta (Δ)” concept. With this approach, it is often assumed that the transition from heavy to very heavy exercise (i.e. the MLSS) occurs at an exercise intensity approximately halfway between the power output corresponding to Th_{la} and the power output corresponding to $\dot{V}\text{O}_{2\text{max}}$ (at 50% Δ). In the present study, the MLSSV occurred at 51 (14)% Δ (range: 33–83% Δ). Therefore, although the group mean suggests that the MLSSV occurs at approximately 50% Δ , the metabolic and gas exchange responses of individual subjects at this exercise intensity may vary considerably depending on whether they are above or below their MLSSV.

The blood [La] at MLSSV was $2.2 (0.1) \text{ mM}$. This was significantly higher than the blood [La] at the Th_{la} [$1.2 (0.6) \text{ mM}$; $P < 0.05$], but appreciably lower than the blood [La] at MLSSV reported previously (Carter et al. 1999; Heck et al. 1985; Jones and Doust 1998). Carter et al. (1999) and Jones and Doust (1998) reported that the blood [La] at MLSSV in subjects of similar fitness to

those involved in the present study was $3.4 (0.6) \text{ mM}$ and $4.4 (0.8) \text{ mM}$, respectively. The reason for this difference is not clear, but it serves to highlight the inter-individual nature of the blood [La] responses at MLSSV. It was of interest that the blood [La] at the MLSSV was not significantly different from that at the LTPV. Furthermore, there were no significant differences between MLSSV and the LTPV for running velocity, $\dot{V}\text{O}_2$ or f_c . This suggests that the LTPV, which can be measured in a single incremental exercise test, might provide a good estimate of the MLSSV. This possibility is worthy of further research attention. However, in the present study, the extent of disagreement between the two variables [bias $\pm 95\%$ limits: $0.3 (1.8) \text{ km}\cdot\text{h}^{-1}$] indicates that estimating MLSSV from LTPV could result in significant over- or underestimation in individual subjects.

Although it is accepted that there are two thresholds in ventilation (the ventilatory threshold and the respiratory compensation threshold) during fast incremental exercise tests, the existence of two thresholds in blood lactate is controversial. Indeed, some groups dispute the existence of threshold phenomena, arguing that the blood [La] response to incremental exercise is continuous (Campbell et al. 1989). Skinner and McLellan (1980) first suggested that two threshold increases in blood [La] occur during incremental exercise. The first Th_{la} represents the first increase in blood [La] above resting values at $\approx 1.0 \text{ mM}$, while the second Th_{la} or lactate turnpoint is characterised by a “sudden and sustained” increase in blood [La] at $\approx 2.0-5.0 \text{ mM}$. Several other groups have identified a lactate turnpoint during incremental exercise (Aunola and Rusko 1992; Davis et al. 1983; Hofmann et al. 1994). Identification of the LTPV was facilitated in the present study by removing the lactate data for the running velocities below the Th_{la} , and plotting the log of [La] against the log of running velocity for the remaining data points. This approach is consistent with the procedures of Hofmann et al. (1994) with the addition of the log-log transformation recommended by Beaver et al. (1985) for the interpretation of blood lactate data. The physiological mechanisms underpinning the increases in blood [La] at Th_{la} and LTP are not clear. However, it is possible that the first increase in blood [La] above baseline during incremental exercise represents an accelerated production of lactate in muscle, whereas the second breakpoint in blood [La] represents the point at which the rate of lactate production outstrips the rate of lactate clearance during prolonged constant-intensity exercise. In other words, continuous exercise at intensities between the Th_{la} and LTP should result in blood [La] values that are elevated above baseline but stable over time. In this respect, the Th_{la} and the LTP represent the boundaries between moderate and heavy exercise, and between heavy and very heavy exercise, respectively.

In conclusion, we found no significant difference between the CV, MLSSV and LTPV during treadmill running across the subject group as a whole, suggesting that the three variables are inter-related and that each

may define the transition between the heavy and very heavy exercise intensity domains. However, close agreement between the variables was not evident in all of the subjects. The bias $\pm 95\%$ limits of agreement suggest that the extent of disagreement between the variables is too great to allow, for example, accurate estimation of the MLSSV from the LTPV in individual subjects. Although we found no significant difference between CV and MLSSV, there was a tendency for CV to overestimate the MLSSV. This would lead to a progressive increase in blood [La] over time if CV were used for training prescription. We therefore recommend that the MLSSV be measured in situations where precision is needed in exercise testing or prescription, and in experimental studies.

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